

A phase-field dataset for dynamic brittle fracture under varying crack configurations and loading

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1. Introduction

Dynamic brittle fracture plays a critical role in the failure of engineering materials and structures subjected to extreme loading conditions, including impact, blast, and high-rate tensile deformation. Accurately predicting crack initiation, propagation, branching, and coalescence remains a major challenge in computational mechanics due to the strongly nonlinear and transient nature of fracture processes [1, 2]. Traditional fracture modeling approaches like Linear elastic fracture mechanics and extended Finite Element Method (XFEM) [3] often require explicit crack tracking, remeshing, or predefined crack paths, making simulations computationally demanding and difficult to generalize to complex crack topologies [4].

Among the different fracture modeling techniques, the phase-field method (PFM) has emerged as a powerful variational framework for simulating fracture evolution in solids [5, 6]. Instead of representing cracks as discrete discontinuities, the method introduces a continuous scalar damage field that smoothly transitions between intact and fractured material states. This formulation captures complex fracture phenomena such as crack nucleation, branching, merging, and arbitrary crack propagation without requiring ad hoc tracking algorithms [7, 8]. Consequently, phase-field formulations have gained widespread adoption in brittle and dynamic fracture simulations. Despite these advantages, dynamic phase-field fracture simulations remain computationally expensive. Resolving rapidly evolving crack fronts requires fine spatial and temporal discretizations, particularly near crack tips where strong gradients develop. Adaptive mesh refinement techniques can significantly improve computational efficiency by refining regions near evolving cracks while maintaining coarser discretizations elsewhere [9, 10, 11]. However, generating large-scale, high-fidelity datasets for dynamic fracture remains a challenging and resource-intensive task.

At the same time, recent advances in machine learning and neural operator methods have created new opportunities for accelerated fracture prediction and surrogate modeling [12, 13, 14]. Data-driven frameworks require high-quality benchmark datasets spanning diverse crack configurations, loading conditions, and fracture behaviors. Publicly available datasets for fracture, particularly those based on high-fidelity phase-field simulations with adaptive meshing, remain limited; to my knowledge, there is no specific dataset for dynamic fracture problems.

To address this gap, we present a comprehensive dataset for dynamic brittle fracture generated using a phase-field fracture framework with adaptive mesh refinement. The dataset includes simulations across varying initial crack locations and loading magnitudes, capturing a broad range of fracture responses, including stable crack growth, branching, and coalescence. For each realization, displacement and phase-field variables are stored throughout the loading history, providing a rich resource for training, benchmarking, and validating machine learning models for fracture prediction. The overall framework for dataset generation and machine learning integration is illustrated in Figure 1. The pipeline is organized into three sequential modules: Module 1 defines the dynamic fracture problem, encompassing geometry, material properties, boundary and initial conditions, loading setup, the phase-field fracture model, and adaptive mesh refinement; Module 2A handles dataset generation by systematically varying crack locations within the domain and loading magnitudes, computing the resulting displacement and damage fields, and storing all outputs at each time step using module 1; Module 2B performs pre-processing of the raw data, including cleaning, normalization, resampling onto a uniform grid, and assembly

of ML-ready datasets; and Module 3 covers the machine learning pipeline, from defining the learning task and model architecture to training, validation, and deployment for fast predictions. In this document, we focus exclusively on Modules 1 and 2A, i.e., the problem formulation and the generation of the high-fidelity simulation dataset. The pre-processing strategy (Module 2B) and the choice of machine learning model (Module 3) are intentionally left open, as these will depend on the specific surrogate modeling task and architecture adopted by the end user.

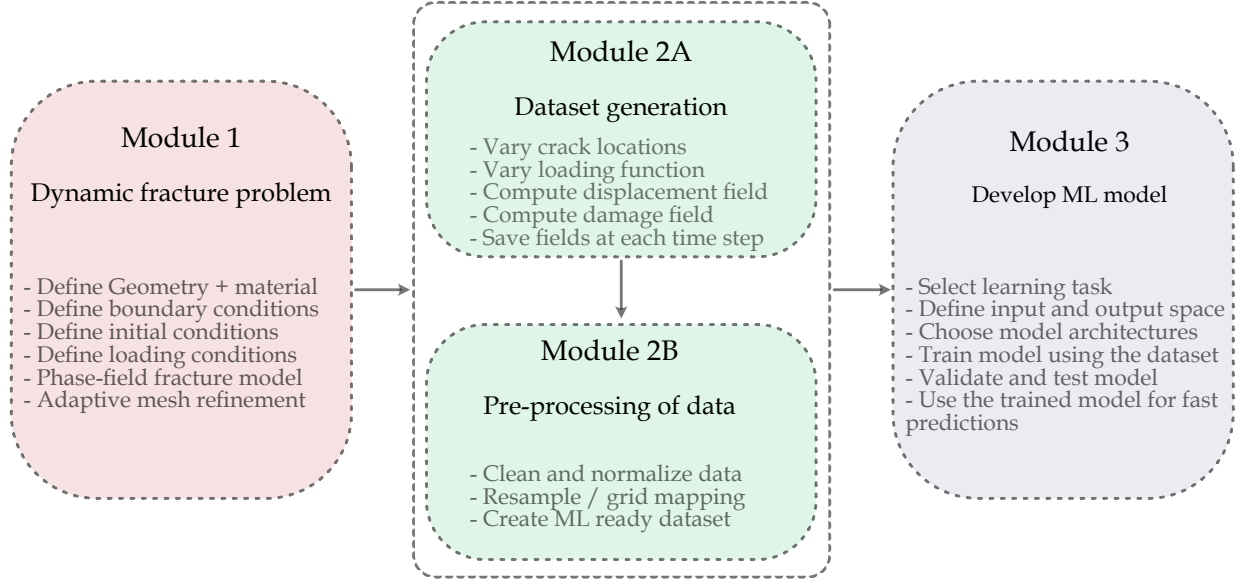


Figure 1: Overall framework for dataset generation and machine learning model development, organized into three sequential modules. Module 1 (Dynamic fracture problem) defines the problem setup, including geometry, material properties, boundary and initial conditions, loading conditions, the phase-field fracture model, and adaptive mesh refinement. Module 2A (Dataset generation) produces the simulation data by varying crack locations and loading functions, computing the displacement and damage fields, and saving all field outputs at each time step. Module 2B (Pre-processing) prepares the raw simulation outputs for machine learning by cleaning and normalizing the data, resampling onto a uniform grid, and assembling the final ML-ready dataset. Module 3 (Develop ML model) covers the full machine learning pipeline: defining the learning task and input/output spaces, selecting and training model architectures on the dataset, validating and testing the trained model, and deploying it for fast fracture predictions.

2. Phase-Field Fracture Formulation

The dataset is based on the standard second-order phase-field formulation for brittle fracture. In this model, fracture is represented by a scalar damage variable $d \in [0, 1]$, where $d = 0$ denotes intact material and $d = 1$ indicates complete fracture. The total system energy is expressed as the Lagrangian functional $L(u, d)$, combining kinetic energy (K), elastic strain energy, fracture dissipation, and external work (P):

$$L(u, d) = K - \int_{\Omega} \psi_s(\varepsilon, d) dV - \int_{\Omega} G_f c_{\alpha} [\ell \alpha(d) + \ell (\nabla d \cdot \nabla d)] dV + P(u), \quad (1)$$

where u denotes the vector-valued displacement field. Here, ℓ is the internal length-scale parameter that controls the width of the diffused crack zone and ensures mesh objectivity. G_f represents the critical energy release rate and is a material property, and $\alpha(d)$ defines the crack geometry function, which is d^2 for AT2 model.

The scaling factor $c_{\alpha} := 4 \int_0^1 \sqrt{\alpha(\beta)} d\beta$ ($c_{\alpha} = 2$, for AT2 model) maintains consistency in the crack density formulation. The kinetic and external work contributions are defined as:

$$K = \frac{1}{2} \int_{\Omega} \rho \dot{u} \cdot \dot{u} dV, \quad P = \int_{\Omega} b \cdot u dV + \int_{\Gamma_t} t \cdot u dA, \quad (2)$$

where ρ is the material density, $\dot{u}$ the velocity, b the body force, and t the applied traction on the Neumann boundary Γ_t . Including kinetic energy allows accurate treatment of inertia-driven fracture events, which are central to dynamic problems. The strain energy density is defined as:

$$\psi_s(\varepsilon, d) = \psi_0^-(\varepsilon) + \omega(d) \psi_0^+(\varepsilon), \quad (3)$$

where $\omega(d)$ is the degradation function satisfying $\omega(0) = 1$ and $\omega(1) = 0$. This ensures that the tensile contribution to the energy vanishes in fully fractured regions, while the compressive part ψ_0^- remains unaffected. Commonly, $\omega(d) = (1 - d)^2$. The spectral decomposition proposed by Miehe *et al.* [15] prevents unphysical crack growth under compression. This variational structure guarantees energy balance and thermodynamic consistency throughout the fracture process. Further model details are provided in Gupta *et al.* [16]. The geometry and boundary condition setup used in the simulations are shown in Figure 2.

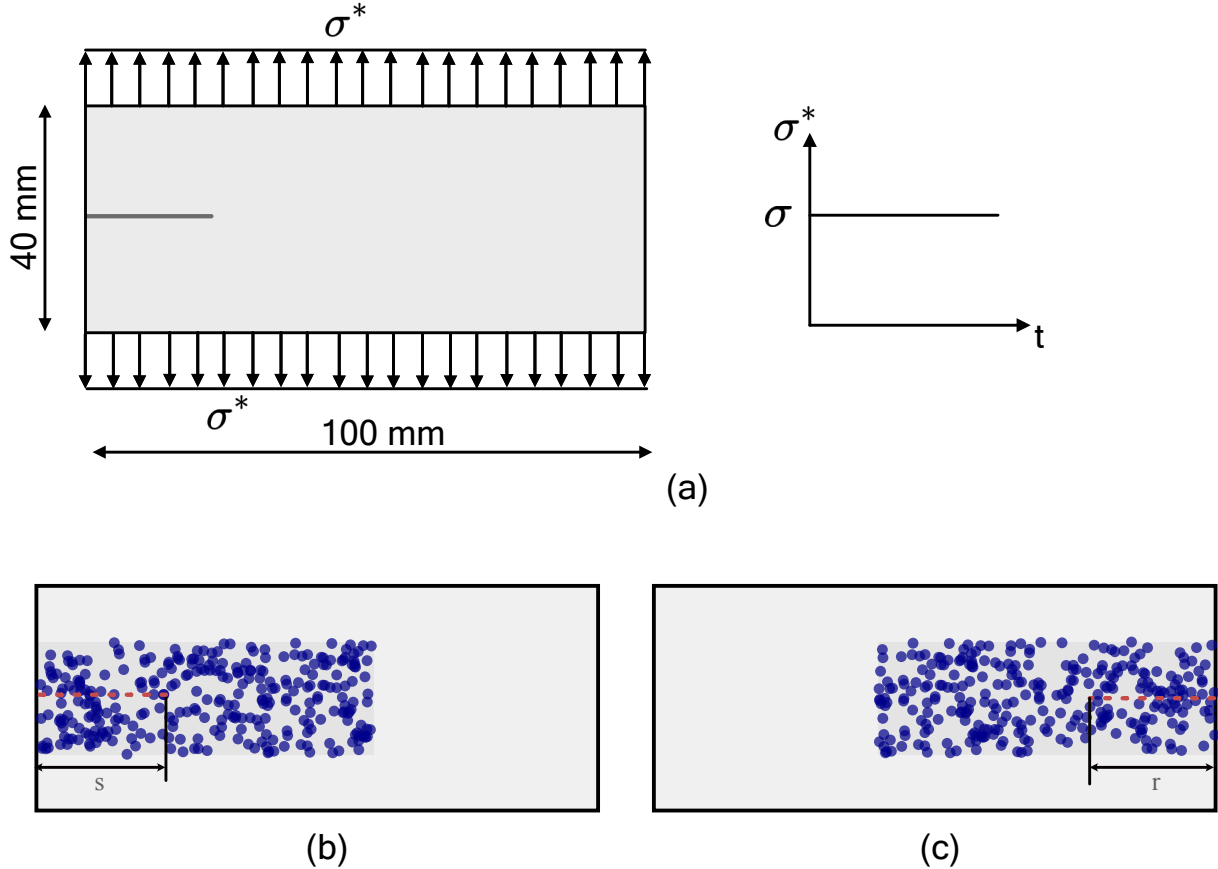


Figure 2: (a) Geometry and boundary condition setup for the dynamic fracture simulations. A rectangular plate of dimensions 100 mm $\times$ 40 mm contains a pre-existing horizontal notch. A uniform tensile traction σ^* is applied simultaneously on the top and bottom boundaries. (b,c) Distribution of initial crack tip positions across the crack configurations in the dataset, where s and r show the initial crack length considered and blue dots show the crack tip locations sampled within the plate domain.

3. Computational framework

The computational setup involves a coupled displacement and phase-field formulation for a rectangular plate of dimensions 100 mm $\times$ 40 mm containing a pre-existing horizontal notch, as shown in Figure 2. A uniform tensile traction σ^* is applied simultaneously on the top and bottom boundaries. Four normalized traction magnitudes are considered: $\sigma^* \in \{1.0, 1.5, 2.0, 3.0\}$, spanning loading regimes from steady crack propagation at lower values to branching and crack coalescence at higher values. The material parameters are held fixed across all

simulations and are provided in Table 1 for reproducibility.

Table 1: Material parameters used in all dynamic fracture simulations.

E_0 [MPa]	ν_0 [-]	ρ [kg/m ³]	G_f [N/mm]	ℓ [mm]
32×10^3	0.2	2450	0.003	0.5

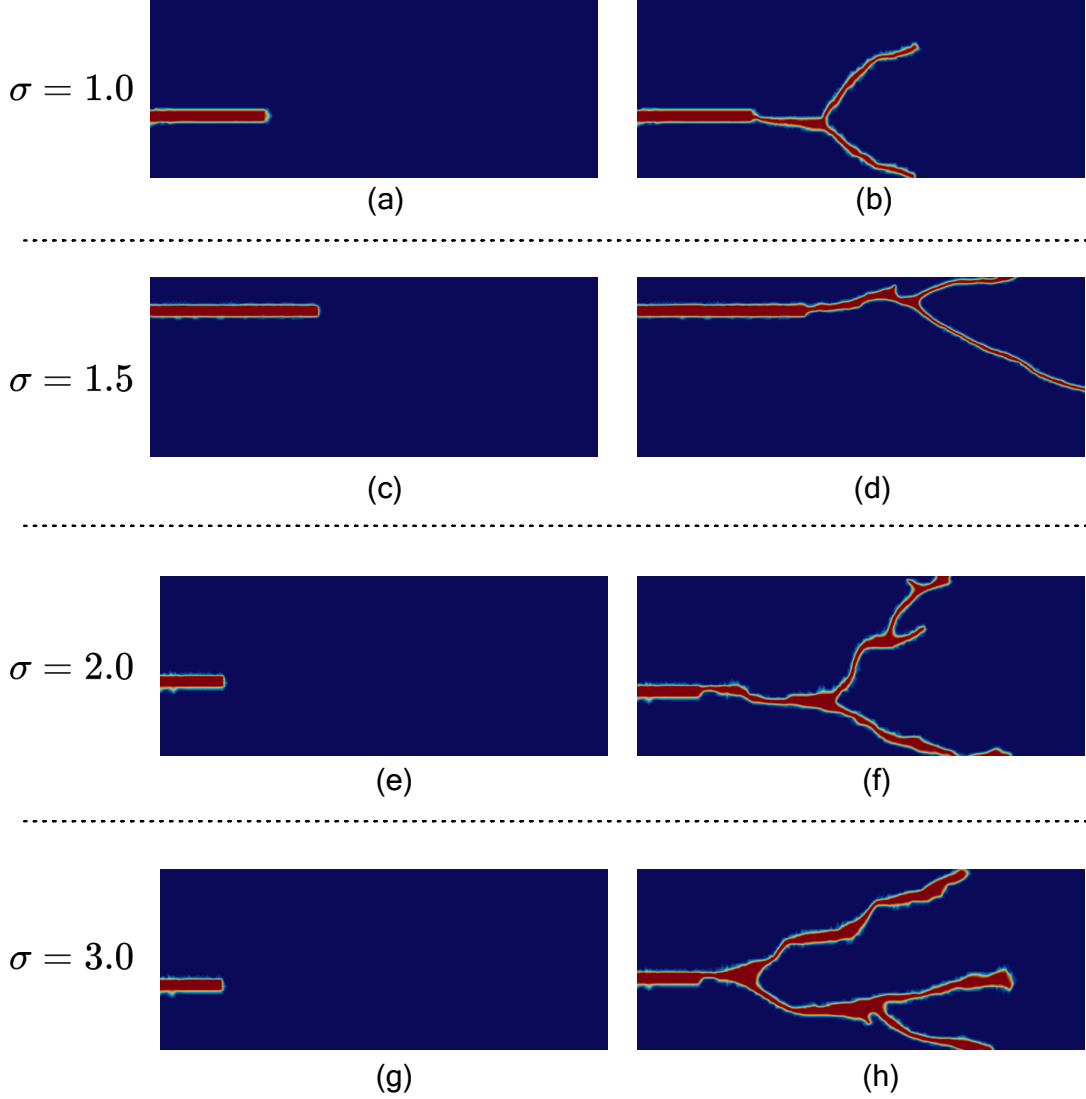


Figure 3: Initial (left column) and final (right column) phase-field crack configurations for a representative sample under each of the four traction magnitudes: $\sigma^* = 1.0$ (a,b), 1.5 (c,d), 2.0 (e,f), and 3.0 (g,h). At lower loading ($\sigma^* = 1.0$), the crack propagates and branches into a single symmetric fork. As the loading magnitude increases, the fracture pattern becomes progressively more complex, exhibiting multiple branches and crack coalescence at $\sigma^* = 2.0$ and 3.0, demonstrating the range of dynamic fracture regimes captured in the dataset.

The variability in the dataset is introduced through two factors: the initial crack position, i.e., as shown in 2 by varying r or s , and the loading magnitude σ^* . The initial crack is a horizontal notch whose left-tip x -coordinate is sampled over $[0, 60]$ mm, right-tip x -coordinate over $[40, 100]$ mm, and y -coordinate over $[12, 37]$ mm. Combined with the four traction magnitudes, this produces a total of **2000 simulations** (500 crack positions $\times$ 4 load levels).

The equations of motion are integrated in time using an implicit Newmark- β scheme. An adaptive mesh

refinement algorithm is used so as to reduce the overall computational time. The simulation is carried out, and the coordinate data, displacement, and phase-field variables are stored after every 5th iteration step, respectively. Figure 3 presents representative phase-field fracture results for each of the four normalized loading magnitudes ($\sigma^* \in \{1.0, 1.5, 2.0, 3.0\}$), comparing the initial notch configuration with the fully evolved crack topology at the final loading step. This range of fracture behaviors from stable propagation to dynamic branching underscores the diversity of the dataset.

4. Dataset description

The dataset is organized as shown in 4, inside a top-level directory `dataset_dynamic/` containing 1,901 phase-field fracture simulations of a 2D specimen under dynamic traction loading, out of 2,000 attempted simulations. For each simulation, three fields are stored:

- Mesh node coordinates (.npz)
- Nodal displacement field (.npz)
- Nodal phase-field damage variable (.npz)

Simulation-level parameters are documented in `metadata.json` at the root of `dataset_dynamic/`, with the following key entries:

- **Material properties:** Young’s modulus $E = 32,000$ MPa, Poisson’s ratio $\nu = 0.2$, density $\rho = 2,450$ kg/m³, tensile strength $f_t = 12$ MPa, and critical energy release rate $G_c = 3 \times 10^{-3}$ kN/mm.
- **Loading parameters:** traction amplitudes at four levels (L1, L1.5, L2, L3) applied on the top and bottom boundaries at $y = 40$ mm and $y = 0$ mm respectively, under two loading directions — Std (initial crack starting from the left side) and Rev (initial crack starting from the right side).
- **Numerical parameters:** phase-field length scale $\ell = 0.5$ mm, Newmark- β parameters $\gamma = 0.5$ and $\beta = 0.25$, staggered scheme tolerance 10^{-6} , and adaptive mesh target $h_{\min} = 0.25$ mm.

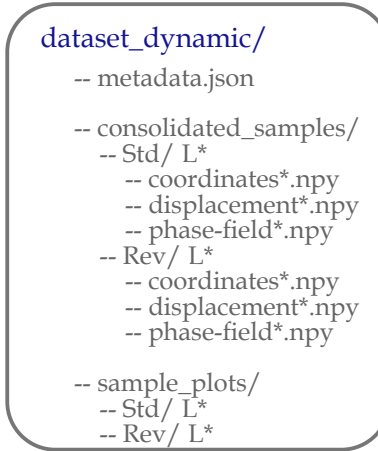


Figure 4: Folder structure of the dataset repository, showing the hierarchical organization by initial crack’s direction (Std, Rev), load level (L1, L1.5, L2, L3), and individual sample directories.

Samples are organized hierarchically inside `consolidated_samples/` by initial crack location and load level (e.g. Std/L1/). Each subfolder contains sequentially numbered directories `sample1/`, `sample2/`, ..., each corresponding to a distinct initial crack location realization. Within each `sample{N}/` directory, fields are stored across three subfolders:

- `coordinates/coord_sample{N}_step{S}.npz`: mesh node coordinates, shape $[n_{\text{nodes}}, 2]$.
- `displacement/disp_sample{N}_step{S}.npz`: nodal displacement (u_x, u_y) , shape $[n_{\text{nodes}}, 2]$.
- `phase_field/phase_sample{N}_step{S}.npz`: phase-field damage ϕ , shape $[n_{\text{nodes}}]$.

Fields are saved every 5 time steps, with $\Delta t = 0.1$, over a maximum simulation time of 80 s. Because adaptive mesh refinement modifies the mesh connectivity during the simulation, coordinates and field values are stored

independently at each saved increment rather than on a fixed mesh topology.

5. Community use

This dataset is made openly available to the research community to facilitate the development, training, and benchmarking of machine learning models for dynamic fracture problems. We encourage its use across a broad range of applications, including the development of surrogate models and neural operators for fast fracture prediction, the benchmarking of existing data-driven architectures against high-fidelity phase-field simulation results, and the exploration of uncertainty quantification and generalization across varying loading conditions and crack configurations. By releasing this dataset, we aim to lower the barrier to entry for researchers working at the intersection of computational mechanics and machine learning, and to promote reproducible, community-driven advances in the simulation and prediction of dynamic brittle fracture.

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